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GLOBAL FACTORS AND EMERGING MARKET SPREADS*

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This article shows that a large fraction of the time variability of emerging market bond spreads is explained by the evolution of global factors such as risk appetite, global liquidity and contagion from systemic events such as the Russian default. This link is robust to the inclusion of country-specific factors and helps to provide accurate long-run predictions. By contrast, changes in credit ratings appear to lag spread movements and elicit little additional effect on the pricing of emerging market debt. The results highlight the critical role played by exogenous factors in the evolution of borrowing costs faced by emerging economies.

Global (exogenous) factors are increasingly seen as key determinants of the borrowing costs of emerging economies, for good reason. In principle, the pricing of debt issued by financially integrated emerging economies should be no different from the pricing of non-investment grade securities in general and low-grade bonds in developed economies in particular. Both should reflect the level of risk of the security and a risk premium (the price of risk) that is, in turn, a reflection of the risk aversion – risk appetite – of the international investor. It follows that variation in emerging market spreads may be driven by exogenous changes in global risk appetite. In addition, global liquidity influences the international cost of capital and, to the extent that this cost affects debt sustainability, emerging market spreads. It follows that an important part of the variability of emerging spreads could be seen as a reflection of exogenous factors (such as the international business cycle) that simultaneously determine both risk appetite and the interest rate.

Indeed, this intuition has not gone unnoticed in the literature. In an early study of emerging market spread determination, Eichengreen and Mody (2000) conclude that 'when it comes to changes in spreads over time, ... these are explained mainly by shifts in market sentiment rather than by shifts in fundamentals'. This contrasts with Dell'Ariccia *et al.* (2002), in a study of moral hazard after the Russian default, who find that emerging market and corporate spreads were significantly but *negatively* correlated after the Russian episode. In turn, IMF (2004) argues that global liquidity (as measured by the 3-month LIBOR) was the most important determinant of the narrowing of emerging market spreads post-September 2001. The link between spreads and global factors has also been highlighted by Grandes (2003), Herrera and Perry (2002), Calvo and Talvi (2004) and García Herrero and Ortiz (2004).

This article examines this premise (namely, that the evolution of emerging market spreads is to a large degree determined by exogenous global factors) in a rigorous and systematic way, with the benefit of a longer time span and a broader set of countries. Specifically, it studies the extent to which changes in international interest rates and

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corporate bond spreads in developed countries explain the dynamic (within) country variability of emerging market spreads in recent years. Based on high frequency (monthly, weekly, and daily) data, the article estimates panel error-correction models of emerging spreads on high-yield spreads and international rates, controlling for the presence of contagion. By contagion we understand cases in which a crisis elsewhere negatively affects an economy with neither real ties with the crisis country nor a visible deterioration in fundamentals (Mexico 1994 and Russia 1998 are the two episodes for which we control in the tests). While the tests also control summarily for time-varying country-specific risk using credit ratings (or, alternatively, country-time dummies), we believe that portraying these coefficients in terms of the explanatory power of domestic factors may be misleading in light of the obvious endogeneity problems – some of which we document in the article.

In addition, these high frequency data lend themselves to an event-study analysis that documents the endogenous nature of credit ratings.

The main empirical findings are the following:

(*i*) Emerging market spreads depend negatively on international risk appetite (as proxied by high-yield corporate spreads in developed markets) and positively on international liquidity (proxied by US Treasury notes). These two exogenous factors explain just over 50% of the long-run time variability of emerging market spreads (14% of the short-run variability) for the period following the Russian default (2000–5), and close to 80% when the elasticities with respect to global factors are allowed to differ across countries.

(*ii*) By contrast, ratings appear to be largely endogenous, reflecting changes in spreads rather than anticipating them. Indeed, a closer look reveals that credit rating or outlook changes lag spread movements and elicit little if any additional effect on spreads.

1. Emerging Market and Corporate Bond Pricing

1.1. *A Reduced-form Model*

Interest rate arbitrage by risk-averse investors implies that

$$(1-q)(1+r)+qV=(1+r_f)+\varphi q \tag{1}$$

where q is the probability of default, V the recovery value after default, r and r_f the interest rates charged on the bond and on a risk-free asset of similar duration, and φ is a parameter that reflects investors' risk aversion. Then, we can express the emerging market spread as:

$$spread_{it} \equiv r_{it} - r_t^f = [\varphi_t + (1+r_t^f) - V_t][q_t/(1-q_t)]\phi_{it} \tag{2}$$

or, more generally, assuming that recovery values are stable over time and comparable across bonds, as

$$spread_{it} = \rho(r_t^f, \varphi_t)\theta[q(X_{it})]\phi_{it} \tag{3}$$

where

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- $\rho(r_t^f, \varphi_t) = [\varphi_t + (1 + r_t^f) - V]$ denotes the price of credit risk, which depends on the international risk-free rate r_t^f and risk aversion φ_t;
- $\theta[q(X_{it})] = \{q(X_{it})/[1 - q(X_{it})]\}$ measures the incidence of the default risk of the issuer, q, itself a function of country-specific (in the case of sovereign debt) or firm-specific (in the case of corporate bonds) fundamentals X_{it}; and
- $\phi_{it} = \phi_i(r_t^f, d_t^c) = (r_t^f)^{\alpha j}\exp(\beta_j d_t^c)$ is a scale factor reflecting the fact that global factors such as global liquidity (measured by the international risk-free rate r_t^f) or episodes of financial distress (represented by the dummy d_t^c) affect corporate and emerging markets differently.

In particular, we have that

$$\ln HY_t = \ln \rho(r_t^f, \varphi_t) + \ln \theta[q(HY_t)] + \alpha_{HY} \ln r_t^f + \beta_{HY} d_t^c \tag{4}$$

where HY_t is a High Yield spread index (the weighted average spread of high-yield corporate debt around the world) and $q(HY_t)$ stands for the associated (weighted average) credit risk, from which we can obtain a good approximation of the log of the price of risk as

$$\ln \rho(r_t^f, \varphi_t) = \ln(HY_t) - \ln \theta[q(HY_t)] - \alpha_{HY} \ln r_t^f - \beta_{HY} d_t^c \tag{5}$$

to characterise emerging market spreads as:

$$\ln embi_{it} = \ln HY_t + \ln \theta[q(embi_{it})] - \ln \theta[q(HY_t)] + \alpha \ln r_t^f + \beta_{HY} d_t^c \tag{6}$$

where $\alpha = \alpha_{embi} - \alpha_{HY}$, and $\beta = \beta_{embi} - \beta_{HY}$. Moreover, assuming for the moment that, in both cases, idiosyncratic credit risk is reasonably captured by the credit rating of the issuer (we relax this assumption below), so that $\ln \theta[qX_{it})] \approx \gamma \ln rating_{it}$ and that $rating_{HY}$ remains relatively stable over time, we obtain the following long-run specification:

$$\ln embi_{it} = \ln HY_t + \alpha \ln r_t^f + \beta_{HY} d_t^c + \gamma \ln rating_{it}. \tag{7}$$

Note that the price of risk captured by the high yield spread is a function of both risk appetite and the risk-free rate. On the other hand, international interest rates may also influence the pricing of sovereign and corporate risk due to international arbitrage: to the extent that it reflects investment returns in the developed world, it is expected to affect negatively the sustainability of emerging market debt and its investor base relative to other risky securities.

There are reasons to expect that the adjustment to changes in its main determinants would not be immediate (for example, imperfect information or market frictions may introduce costs that require that deviations from the long-run level exceed a minimum – possibly asymmetric – threshold to trigger a price adjustment). While we are primarily concerned about the determination of emerging market spreads in the long run, to provide a complete characterisation of the impact of global factors, we also examine the reaction to short-run deviations, and the speed of convergence to the long-run level, by augmenting the previous long-run specification with an error correction equation.

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Table 1

Summary Statistics

Variable	Frequency	*N*	Mean	Median	Std Dev	Min	Max
(Monthly sample)							
embi	Monthly	3,309	648.1935	402.7520	831.5839	7.0240	7,078
HY	Monthly	191	580.8115	523	203.5349	307	1,080
rating	Monthly	4,351	10.8322	11	3.3172	1	18
10YT	Monthly	191	5.9228	5.86	1.3511	3.37	9.04
VIX	Monthly	191	19.4612	18.88	6.3882	10.63	44.28
rating + outlook	Monthly	4,283	10.8453	11	3.3537	1	18
Rating (*HY*)	Monthly	131	4.4352	4.3925	0.1323	4.2347	4.7403
HY (*Split BBB*)	Monthly	131	250.8441	224.88	103.8955	114.25	563
(Weekly sample)							
embi	Weekly	14,370	645.7757	399	832.7009	−6	7,222
HY	Weekly	493	583.6755	556	211.4735	301	1,116
rating	Weekly	18,919	10.8375	11	3.3125	1	18
10YT	Weekly	834	5.9284	5.88	1.3443	3.18	9.07
VIX	Weekly	836	19.4045	18.36	6.4250	9.48	45.74
rating + outlook	Weekly	18,620	10.8489	11	3.3506	1	18

1.2. *The Data*

To proxy for the price of risk, we use Credit Swiss First Boston's High Yield Index (*HY*), which measures the spread over the US treasuries yield curve at the redemption date with the worst yield – an alternative measure of high yield spreads prepared by J. P. Morgan yielded almost identical results. Emerging market sovereign spreads are measured as the spread over Treasuries of J. P. Morgan's EMBI_Global index (*embi*) for each of the 33 emerging economies included in the Global portfolio (period coverage varies across countries, as reported in Table A2 in the Appendix). The credit rating variable (*rating*) is constructed based on Standard & Poor's rating for long-term debt in foreign currency. As a proxy for international liquidity, we use the 10-year US Treasury rate (*10YT*), although we also run tests using the US$ and the DM/Euro 6-month LIBOR for robustness (sourced from the Federal Reserve and the Bundesbank, respectively). In all cases, we work alternatively with monthly, weekly and (occasionally) daily data. Table 1 shows summary statistics of the variables used in this study (Table A1 in the Appendix shows variable definitions and sources).

1.3. *The Methodology*

The econometric model used in this article describes a basic long-run relationship between the market spreads, the high yield index and the international rate. Standard unit root tests, such as Maddala and Wu (1999) and Choi (2001), reveal that all variables have a panel unit root individually, both at the monthly and weekly frequencies (González-Rozada and Levy Yeyati, 2006). Since there is evidence of cointegration, by the Granger Representation Theorem the variables in the long-run equilibrium relationship have a panel error correction representation (PECM), which we estimate using the Engle-Granger methodology (Engle and Granger, 1987), a two-stage modelling strategy that may be formalised as follows.

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In stage one, we estimate the long-run parameters of the cointegration equation

$$\ln(embi_{it}) = \alpha_{0i} + \alpha_1 \ln(HY_t) + \alpha_2 \ln(r_t^f) + \alpha_3 \ln(d_t^c) + \alpha_4 \ln(rating_{it}) + \varepsilon_{it}, \tag{8}$$

from which we get the residuals ($\check{z}_{i,t}$). In stage two, we use the error correction term lagged once, $\check{z}_{i,t-1}$, and estimate a PECM to get the short-run dynamics. Because we do not know the lag structure of the short-term dynamics out of the steady state, we begin with a general specification of the form

$$\Delta \ln(embi_{i,t}) = \gamma_{0i} + \gamma_1 \check{z}_{i,t-1} + \sum_{j=0}^{p} \Gamma_{0j} \Delta \ln(HY_{t-j}) + \sum_{j=0}^{q} \Gamma_{1j} \Delta \ln(r_{t-j}^f) + \sum_{j=0}^{m} \Gamma_{2j} \Delta \ln(W_{t-j}) + \\ + \sum_{j=0}^{n} \Gamma_{3j} \Delta \ln(rating_{i,t-j}) + \sum_{j=1}^{r} \Gamma_{4j} \Delta \ln(embi_{i,t-j}) + u_{it}. \tag{9}$$

We estimate both equations using the least squared dummy variable (LSDV) approach. The time dimension in our datasets is large enough that the LSDV estimation bias goes to zero. However, in order to check the robustness of our estimations, we also estimated (8) using a panel dynamic ordinary least squares procedure and (9) using a bias corrected least squared dummy variable procedure, finding similar results to those presented here.

2. Empirical Results

At first sight, the data appear to support the view that global factors exert a strong influence on emerging market spreads. Figure 1 illustrates the point, presenting the median of individual EMBI spreads at each point in time and the distribution around the cross-section mean (all variables normalised by their country-specific means), together with our two main proxies of global factors. The composite EMBI index typically used as a crude measure of emerging market risk pools together individual country portfolios of diverse duration and risk, and may be too sensitive to big outliers (highly indebted countries undergoing a crisis). Hence, our focus is on the distribution. A simple comparison of the first two panels shows that high yield spreads in developed and developing countries (*HY* and *embi*) have indeed moved together over the last 10 years – somewhat less so in the earlier period due to the strong impact of contagion from the Mexican and Russian crises. Moreover, there seems to be also a direct relationship between the emerging markets spreads and international liquidity, measured through the US Treasury notes, 10 year constant maturity yield, as shown by a statistically significant correlation coefficient of about 0.12.

This first impression is strongly confirmed by estimates that are more rigorous. Table 2 reports our baseline specification and our main results. Column 1 shows the estimated long-run equation of the spread as a function of the high yield index, the international interest rate, contagion from crises abroad and the credit rating, using monthly data. As can be seen, the correlation is large and highly significant for all the variables. In particular, two exogenous global factors *HY* and *10YT* explain about 30% of the dynamic (within) variability of spreads, (37% if we add contagion to the list), while the inclusion of credit ratings brings this number to close to 60%. The short-run

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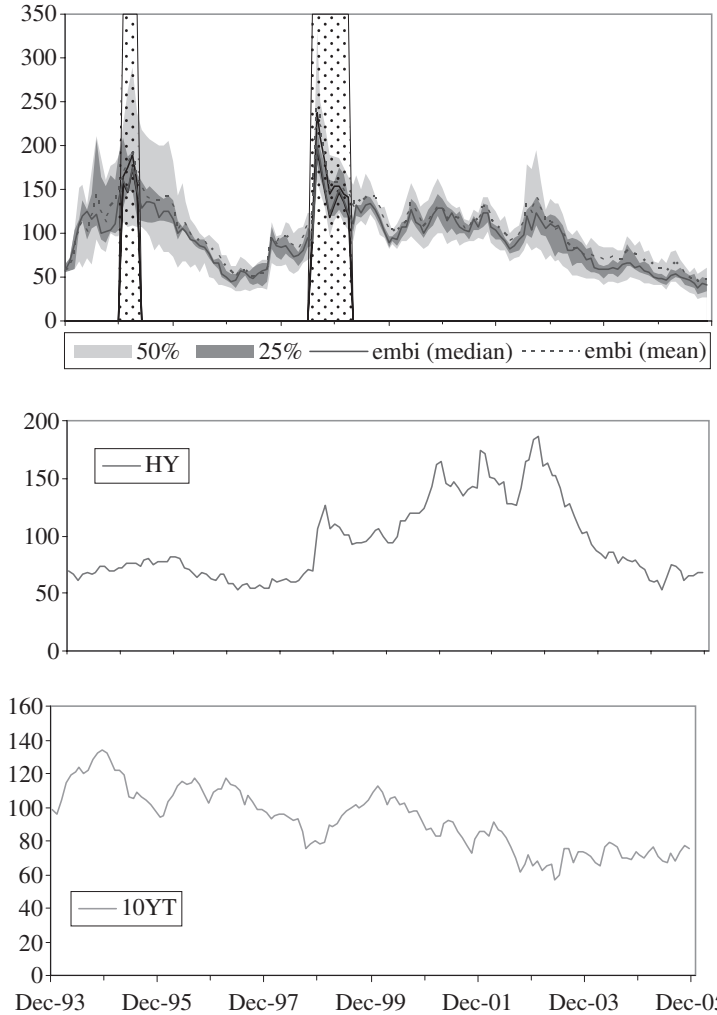

Fig. 1. *HY Index and Embi: Mean Normalised Values* (Mean = 100)

equation is also consistent with our priors (column 2). All variables have a strong contemporaneous impact on spreads but no delayed effect, and roughly 7% of deviations from the long-run level are eliminated per month, so that the average lag length is about fourteen months. Weekly data tell the same story (columns 3 and 4): high correlation, strong explanatory power (about 60% of dynamic variability) and slow convergence (roughly 33 weeks). These results proved to be robust to a number of alternative specifications, including the distinction between positive and negative changes in global factors and the use of the *VIX* (a measure of the volatility implied in the pricing of options on US stocks compiled by the Chicago Board Options Exchange) as an alternative proxy for risk aversion (González-Rozada and Levy Yeyati, 2006).

Note that the influence of the international interest rate goes beyond the standard arbitrage view that indicates that portfolio flows respond to increases in the

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Table 2

Global Factors and Emerging Market Spreads

	Monthly		Weekly	
	(1)	(2)	(3)	(4)
$\ln(HY_t)$	0.776		0.773	
	(10.84)***		(9.41)***	
$\ln(rating_{i,t})$	−1.527		−1.537	
	(8.13)***		(7.96)***	
$\ln(10YT_t)$	1.094		0.964	
	(8.09)***		(7.07)***	
Contagion Russia	0.610		0.568	
	(8.03)***		(4.36)***	
Contagion Mexico	0.186	0.045		
	(2.20)**	(4.58)***		
$EC_{i,t-1}$		−0.070		−0.030
		(5.34)***		(3.25)***
$\Delta\ln(HY_t)$		1.108		0.834
		(6.02)***		(6.52)***
$\Delta\ln(rating_{i,t})$		−0.637		−0.429
		(3.22)***		(3.76)***
$\Delta\ln(10YT_t)$		0.272		0.134
		(1.81)*		(1.27)
$\Delta\ln(embi_{i,t-1})$		−0.160		−0.411
		(4.69)***		(8.41)***
$\Delta\ln(embi_{i,t-2})$				−0.191
				(5.04)***
Constant	2.546	−0.007	2.774	−0.003
	(3.22)***	(0.94)	(3.31)***	(1.00)
Observations	2,767	2,689	11,141	10,983
R-squared (within)	0.5826	0.2853	0.5918	0.3152
% variance explained by HY_t	17.0428	21.8848	23.4124	15.1797
% variance explained by HY_t and $10YT_t$	31.3530	22.5379	32.6942	15.2855
% variance explained by HY_t, $10YT_t$ and *Contagion*	37.0128	22.5667	37.8266	

Robust t statistics in parentheses, clustered by year.
* Significant at 10%; ** significant at 5%; *** significant at 1%.

international rate (Calvo *et al.*, 1993). Indeed, the evidence that sovereign spreads adjust close to one to one to changes in the foreign rate implies that borrowing costs in emerging economies respond more than proportionally to the interest rate cycle in the developed world (the USD and the DM/Euro LIBOR, both correlated with *10YT*, yield comparable results omitted here for conciseness and available from the authors on request).

2.1. *Global Factors Then and Now*

A natural question regarding the connection between bond pricing in developed and developing countries is whether it changed (and, in particular, strengthened) over the years, due to the growing familiarity with the emerging market asset and the increasing integration of capital markets (and bondholders). We analyse this aspect in Table 3, where we split the monthly sample into two sub-samples (1993–9 and 2000–5) and rerun the regressions in columns (1)–(2) in Table 2. Standard tests reveal the presence of a structural break around that date, see González-Rozada and Levy Yeyati (2006).

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Table 3

Global Factors and Emerging Market Spreads Over Time

	Full (unbalanced) sample						Argentina, Brazil & Mexico (balanced sample)			
	1993–9		2000–5		1993–2005		1993–99		2000–5	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$\ln(HY_t)$	1.032		0.899		0.991		1.266		0.966	
	(3.80)***		(22.68)***		(19.02)***		(2.76)***		(10.95)***	
$\ln(HY_t) \times d_{00-05}$					−0.065					
					(5.05)***					
$\ln(rating_{i,t})$	−2.177		−1.268		−1.501		−1.915		−0.659	
	(4.89)***		(7.00)***		(8.69)***		(3.70)***		(6.56)***	
$\ln(10YT_t)$	−0.547		0.761		0.571		0.589		0.072	
	(0.83)		(7.19)***		(2.98)***		(2.32)**		(0.40)	
Contagion Russia	0.043				0.290		0.062			
	(0.35)				(4.87)***		(0.29)			
Contagion Mexico	0.336	0.058			0.159	0.045	0.333	0.111		
	(2.15)**	(3.02)***			(1.90)*	(4.80)***	(3.96)***	(8.22)***		
$EC_{i,t-1}$		−0.093		−0.125		−0.078		−0.149		−0.166
		(4.86)***		(3.85)***		(6.30)***		(1.67)*		(4.02)***
$\Delta\ln(HY_t)$		1.617		0.827		1.534		1.836		0.896
		(31.85)***		(8.19)***		(15.72)***		(14.28)***		(4.23)***
$\Delta\ln(HY_t) \times d_{00-05}$						−0.676				
						(5.17)***				
$\Delta\ln(rating_{i,t})$		−1.503		−0.435		−0.626		−0.869		−0.412
		(4.97)***		(4.18)***		(3.20)***		(1.05)		(3.93)***
$\Delta\ln(10YT_t)$		0.431		0.152		0.184		1.302		0.099
		(1.28)		(0.88)		(1.41)		(3.34)***		(0.33)
$\Delta\ln(embi_{i,t-1})$		−0.138		−0.158		−0.163		−0.089		−0.053
		(2.62)***		(3.10)***		(4.70)***		(3.03)***		(0.54)
Constant	5.666	−0.011	1.608	−0.009	2.271	−0.009	1.971	−0.012	1.302	−0.005
	(2.07)**	(0.73)	(4.62)***	(1.33)	(4.11)***	(1.65)*	(0.54)	(0.71)	(2.45)**	(0.45)
Observations	890	852	1,877	1,816	2,767	2,689	203	197	170	165
R-squared (within)	0.5838	0.4535	0.7083	0.2318	0.6247	0.3090	0.5955	0.4587	0.7917	0.3285
% variance explained by *HY*	37.6916	35.3900	46.5491	13.9484	37.5119	24.2838	38.1353	30.9003	73.4962	21.1277
% var. explained by *HY* and *10YT*	38.1161	37.7223	51.9851	14.0062	40.0107	24.6459	48.1206	37.7439	73.5161	21.1931
% var. explained by *HY*, *10YT* and *Contagion*	40.5705	37.8838			41.0670	24.6727	52.6127	38.8135		

Robust t statistics in parentheses, clustered by year. * Significant at 10%; ** significant at 5%; *** significant at 1%.

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The results, presented in columns (1) to (4), clearly show that, while the *HY* coefficients (both in the short and long-run equation) are larger for the earlier period, their explanatory power is comparable (and increases for the long-run equation) in the later years. This could reflect the fact that high yield spreads were more volatile in the late period. In what follows, our tests are estimated based on the 2000–5 sub-sample. These findings are confirmed when we replicate the exercise for the full sample and control the effect on the later period using an interaction of the *HY* variable with a 2000–2005 period dummy (d_{00-05}) in columns 5 and 6.

Given that emerging economies entered the *EMBI* portfolio at different points in time, it is natural to ask whether the parameter change documented above is simply due to the combination of cross-country differences and a changing sample composition. To dispel these doubts, we replicate columns (1) to (4) for a balanced sample, focusing on three Latin American countries that have been in the emerging market class since the very beginning and that have historically represented a large portion of the whole emerging market portfolio: Argentina, Brazil and Mexico (see columns 7 to 10). The messages for this balanced sample are consistent with the previous one: a strong impact of global factors throughout the sample, with *HY* exhibiting increasing explanatory power over the years. At any rate, while the incidence of global factors is not specific of a particular time, in light of the previous results, in what follows we restrict our attention on the later period.

Are the reported effects economically important? A quick computation based on the estimates for the later period (Table 3, column 3) indicates that they are. A 10% increase in the risk-free rate yields a 7.6% increase in spreads, while a similar change in the high yield spread is reflected almost one to one on emerging market spreads (the elasticity is 90%). On the other hand, a one-notch upgrade in ratings (approximately a 10% increase in the rating dummy while measured at the sample mean), is associated with a reduction in spreads of 35 bps. Thus, ignoring endogeneity problems and taking the ratings coefficient at face value, these results would indicate that, based on the values of global factors and the emerging market average rating by end-2005 (BB+, roughly equal to the mean for the whole sample), an upgrade to BBB− could be undone by an increase in the risk-free rate of 70 bps or a rise in the high yield corporate spread of 50 bps. Conversely, the effect of 100 bps increases in both global factors – variations that are relatively common for these two variables – would exceed that of a three-notch downgrade.

2.2. *Missing Fundamentals?*

In the previous tests, credit ratings were treated mainly as a control for country-specific fundamentals, focusing the analysis on the results associated with risk. In so doing, we abstracted from the influence of country fundamentals *per se*, beyond what is captured by the rating assigned to the country. There are at least two reasons why one would like to have a closer look at the role of actual fundamentals. The first one is the belief that ratings do not always reflect the macroeconomic context. More precisely, many observers have pointed out that ratings provide, at best, only a partial account of the actual likelihood of default of individual countries. In fact, there is a growing belief that rating agencies tend to lag spreads in their reaction to significant news and, more

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generally, to reflect credit risk only imperfectly – hence, the substantial dispersion of spreads within the same rating category. At any rate, to the extent that ratings are only a partial proxy for country-specific factors, it is essential for our purposes to test whether the incidence of global factors reported above is robust to a more parsimonious specification.

A second, related motivation is to evaluate the influence of ratings beyond and above the evolution of country-specific fundamentals. While their information content has been questioned on various grounds, the results reported above indicate that they exhibit a significant explanatory power for both the long-run level and the short-run variation of emerging spreads. Is this result indicating that they adequately capture relevant country information, or that they exert an influence of its own, not necessarily related to the evolution of the country's economy? In other words, could ratings be considered as an additional exogenous factor that influences the borrowing cost of emerging economies, independently of whether they reveal valuable information? If so, the influence of ratings on bond pricing could be regarded as an additional external source of volatility, bearing the question about the extent to which markets react to the relevant country-specific economic data. However, Mora (2004), in an updated assessment of this issue, emphasises that the lagging nature of ratings may actually smooth out the impact of deteriorating fundamentals in the run up to a crisis.

These considerations presume that actual fundamentals may influence both the level of spreads and the way they commove with risk appetite beyond what is summarised by the credit rating. To verify this view, we add to our long-run specification dummies per country-year (in regressions using monthly data) and country-month (in regressions using weekly data) to capture the influence of fundamentals identified in the literature as determinants of sovereign risk, such as the country's leverage ratio, the degree of (financial and institutional) development, or cyclical output fluctuations, which are typically sampled at those frequencies.

Table 4 reports the results. The coefficients and explanatory power of the original baseline equation variables remain notably stable using monthly, weekly and even daily datasets, indicating that their influence is largely independent of a country's fundamentals, although the power of the international rate is weakened by the inclusion of country-time dummies. Note that, under the conservative assumption that all the additional explanatory power resulting from the inclusion of country-time dummies can be attributed to country-specific time-varying factors, global factors would still account for more than half of time variability in all cases. However, this would overstate the explanatory power of country-specific fundamentals if the country dummies were to capture in part the differential effect that changes in global factors have on each individual countries. Note that this differential effect should be reflected in the residuals at the country-time dimension and thus be partially picked up by the dummies.

This intuition is confirmed in columns (3), (6) and (9), where we combine the global factors with country dummies in order to allow the elasticities with respect to changes in global factors to vary across countries. As can be seen, while the average coefficient is close to the one found in the original specification, the explanatory power of both global factors combined jumps from about 50% to close to 80%, at the expense of the country-time dummies. Irrespective of whether the latter truly capture the

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Table 4

Missing Fundamentals? (2000–2005)

	Monthly			Weekly			Daily		
			Country-specific elasticities†			Country-specific elasticities†			Country-specific elasticities†
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\ln(HY_t)$	0.819	0.789	0.773	0.809	0.746	0.758	0.832	0.733	0.732
	(22.80)***	(29.54)***		(26.41)***	(6.08)***		(27.33)***	(10.08)***	
$\ln(rating_{k,t})$	−1.720	−1.302	−1.071	−1.732	−0.209	−0.197	−1.725	−0.318	−0.309
	(8.83)***	(11.75)***	(6.41)***	(8.94)***	(2.53)**	(2.21)**	(8.84)***	(2.88)***	(2.67)**
$\ln(10YT_t)$	0.779	0.301	0.262	0.800	−0.025	−0.059	0.845	−0.081	−0.261
	(7.23)***	(3.47)***		(7.34)***	(0.14)		(8.30)***	(0.76)	
Constant	3.150	3.229	2.777	3.204	1.550	1.793	−0.906	2.390	2.666
	(7.37)***	(7.47)***	(4.15)***	(7.74)***	(1.37)	(1.59)	(2.23)**	(2.05)**	(2.37)***
Observations	1,747	1,747	1,747	7,650	7,650	7,650	36,658	36,658	36,658
% Var. exp. by *HY* and *10YT*	0.5190	0.5190	0.7970	0.5069	0.5069	0.7853	0.5488	0.5488	0.7662
% Var. exp. by ctry-time dummies		0.3615	0.1068		0.3045	0.1165		0.2888	0.1597
R-squared (within)	0.7170	0.9018	0.9155	0.6960	0.8856	0.9219	0.7151	0.9041	0.9324
Fixed effects	ctry	ctry-year	ctry-year	ctry	ctry-month	ctry-month	ctry	ctry-month	ctry-month

Robust t statistics in parentheses, clustered by year.
* Significant at 10%; ** significant at 5%; *** significant at 1%.
† Global factors $\mathrm{Ln}(HY_t)$ and $\mathrm{Ln}(10YT_t)$ are interacted with country-specific dummies to obtain country-specific elasticities. The Table reports the average for ease of comparison.

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incidence of domestic factors, the evidence on the explanatory power of global factors appears to be remarkably strong.

2.3. *Ratings*

While the robustness of the results is reassuring for the two exogenous global factors, it is somewhat intriguing for the case of ratings, which are, in principle, conceived as summary indicators of the relevant country-specific factors now included. As noted, this may be due to the fact that, although investors generally recognise the limitations that ratings display in practice, the norms that inform their decisions force them to take credit ratings as an additional argument – suggesting that, to the extent that they reflect the evolution of the country's economy only partially, they may be regarded as a source of variability that is partially exogenous to the policy maker.

The strong empirical association between ratings and spreads has typically been interpreted as evidence that ratings have an impact on the cost of borrowing. For example, Kaminsky and Schmukler (2002) use panel and event studies to show that changes in ratings and outlooks are positively correlated with emerging market spreads, particularly so during crisis periods. Interestingly, while they note that upgrades occur when markets are rallying and (more visibly) downgrades are affected when they are collapsing, they conclude that ratings are a *cause* of instability during crises times, acting as a signal that coordinates investors towards a bad equilibrium. Similarly, while Reisen and von Maltzan (1998) find evidence of a two-way Granger causality for a sample of developed and emerging countries over the 1988–95 period, they conclude, using event studies, that negative rating news elicit a significant impact on spreads.

However, doubts about the direction of causality have been mounting over the years. For example, Cantor and Packer (1996) find that one variable (GDP *per capita*) explains 80% of the cross-country variance in credit ratings, at odds with the forward-looking, medium-term qualitative and quantitative assessments that one would expect from the rating agencies. More to the point of the present article, the ability of rating agencies to reflect country fundamentals dynamically has been the object of much discussion, particularly in the context of the various debt crises of the late 1990s and early 2000s. Specifically, rating agencies have been criticised for overreacting (in the case of Korea; see Reisen and von Maltzan, 1998; Huhne, 1998) after lagging behind the facts – in the context of the Asian crises (IMF, 1999) and, more recently, the Uruguayan crisis, when they preserved the country's investment grade until early 2002, when the systemic crisis was well underway. In the same vein, it has been argued that they have been excessively inertial (thus involuntarily stabilising the markets in the run up to a crisis; Mora, 2004) as well as excessively pro-cyclical (contributing to the collapse by downgrading a country by several notches once the crisis is inevitable; Ferri *et al.*, 1999). In addition, studies of the determinants of currency crises finds no significant link between recent changes in credit ratings and crisis propensity (Reinhart, 2002), confirming the suspicion that ratings fail to anticipate developments in the country's solvency with a reasonable lag. Table 5 summarily illustrates the point: 12 months before recent default episodes, sovereign ratings assigned by the main rating agencies failed in many cases to sound the alarm.

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Table 5

Sovereign Credit Ratings 12 Months before Default

Country	Month of default	S&P		Moody's	
Dominican Republic	February 2005	CC	2	B3	6
Venezuela	January 2005	B–	6	Caa1	5
Uruguay	May 2003	BB–	9	Ba2	10
Indonesia	April 2002	B–	6	B3	6
Argentina	November 2001	BB–	9	B1	8
Ukraine	January 2000	n/a	n/a	B3	6
Indonesia	April 2000	CCC+	5	B3	6
Ecuador	July 2000	n/a	n/a	B3	6
Pakistan	January 1999	B+	8	B2	7
Russian Federation	January 1999	BB–	9	Ba2	10

Source. Authors' calculations based on S&P and Moody's data.
Foreign currency rating of long-term debt. 1 = Default, 13 = high grade.

Put together, this body of work suggests another, simpler hypothesis to explain the robustness of the spread-rating link to the inclusion of country-time dummies: ratings are, in most cases, endogenous to spreads. Figure 2 illustrates this point graphically. Mimicking an event-study exercise, the first panel shows the residuals from a regression

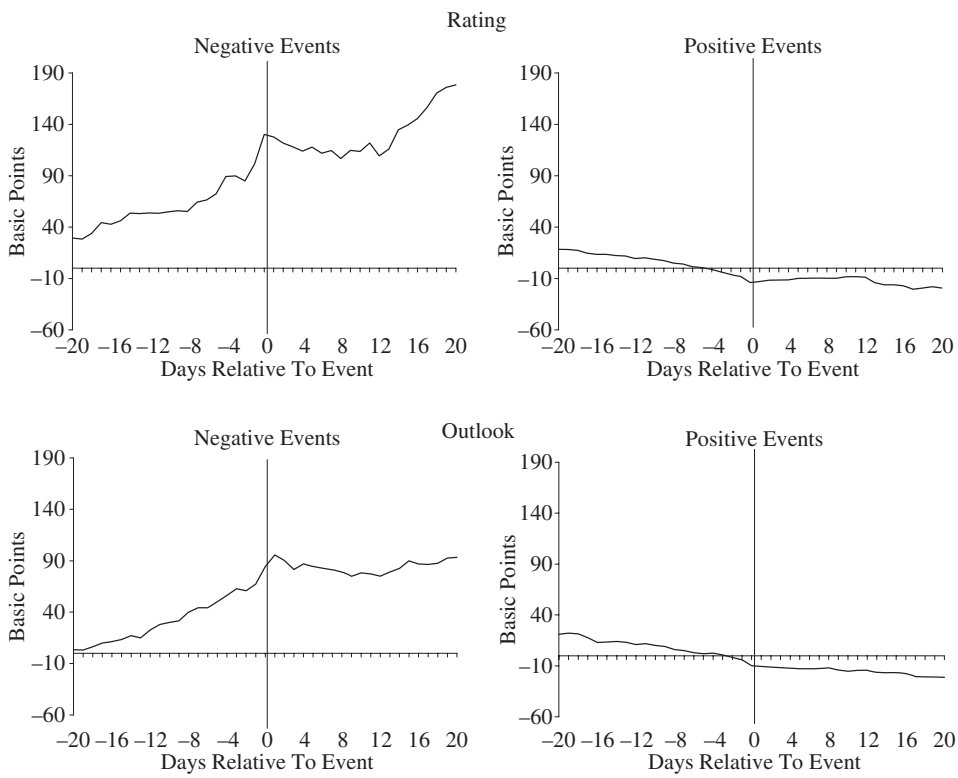

Fig. 2. *Robustness IV, Events Study: Rating and Outlook Changes*

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using the specification of column (3) of Table 3 without the ratings variable, averaged over a 40-day window centred on the grading event, where the latter coincides with positive and negative changes in the rating. As the Figure clearly shows, downgrades are *preceded* by increases in spreads and, apart from a mild contemporaneous adjustment (of about 50 bps), exert no substantial impact. The opposite applies to upgrades, although in this case the preceding decline in spreads is smaller.

Does this evidence prove that ratings are endogenous to market reaction (as reflected by spreads)? Not necessarily, since the market may be simply reacting in anticipation of a change in ratings. Moreover, agencies themselves typically anticipate regarding by changing their own credit outlook – a reason to use the latter to refine the information contained in the rating. However, the second panel of the Figure casts doubt on this possibility. There, we replicate the event-study exercise, this time defining as an event a change in the credit outlook given by the rating agency. The outlook could be thought of as a five-notch grading scale centred on the credit rating: positive, positive watch, neutral, negative watch and negative. The logic is straightforward: if the rating does more than just validate the perceptions of the market, then changes in outlook (the way agencies have to signal the presence of new information that may merit a risk reassessment) should have an impact on asset prices. The results, however, are strikingly similar to those in the first panel. In line with these findings, while adjusting the ratings to reflect changes in the credit outlook improves their explanatory power slightly, it does not introduce any visible change in the remaining coefficients (González-Rozada and Levy Yeyati, 2006).

In sum, the presumption that ratings are a reasonable proxy for fundamental risk is questioned by the data. While their inclusion as a control may still be justified (since they appear to exert a contemporaneous influences on spreads and, at any rate, their endogeneity should not bias the estimation of the remaining coefficients), attributing the strong link between ratings and spreads to the incidence of country-specific factors may be misleading, overstating the role of the latter – and understating the influence of global factors.

2.4. *Predictions*

One way to gauge how much of the variation in emerging market spread can be explained by the few exogenous variables identified in our baseline model consists in simulating the path of individual spreads based solely on the long-run specification. We perform this test by calibrating this equation using data through to end-2001, and simulating the behaviour of spreads for the remaining period (January 2002–November 2005). As a result, some countries that were included in the EMBI after (or shortly before) 2001 are dropped due to lack of observations. Specifically, in order to assess the relative predictive power of global factors, we re-estimate the long-run specification of column (3) of Table 3 and compute the out-of-sample path that results from variations in the global variables *HY* and the international rate, keeping ratings fixed at their end-2001 levels.

Figure 3 reports the results (naturally, one could readily obtain much more precise one-period-ahead forecasts by estimating the short-run equation and adding it to the long-run one, i.e., computing $\mathrm{E}[\ln(embi)_t] = \mathrm{E}[\ln(embi_{\mathrm{long},-1}] + \mathrm{E}[\Delta\ln(embi)]$. Since

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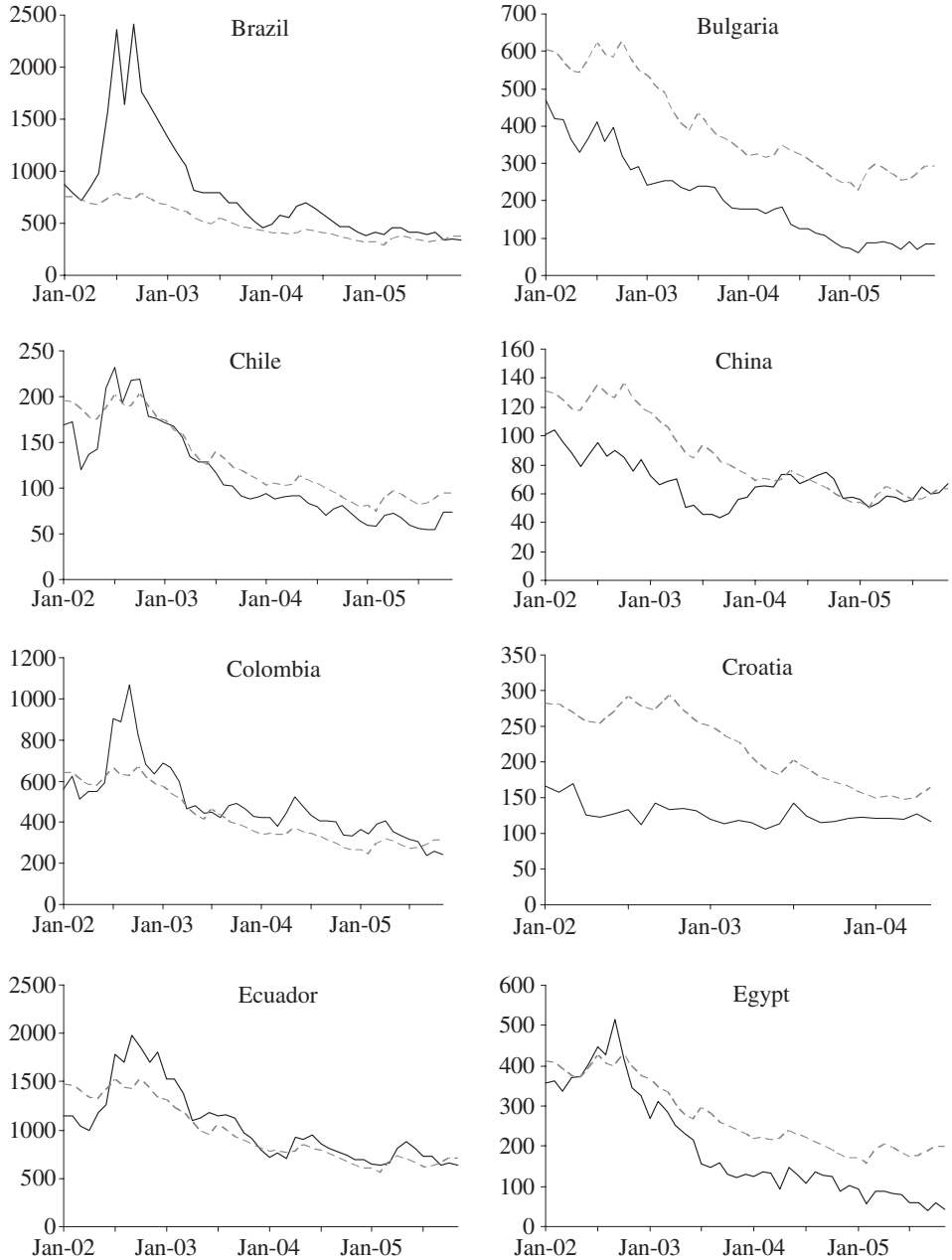

Fig. 3. *Spreads: Out-of-sample Forecasts*
Note: The solid line indicates the EMBI data; the dotted line indicates predictions using HY and 10YT.

our interest is in the *persistent* effect of global factors, the results are omitted here but they are available from the authors on request). Figure 3 compares the actual EMBI spread with the predictions from the long-run equation. Three aspects deserve to be stressed in the Figures. First, predictions are generally quite good, including for many

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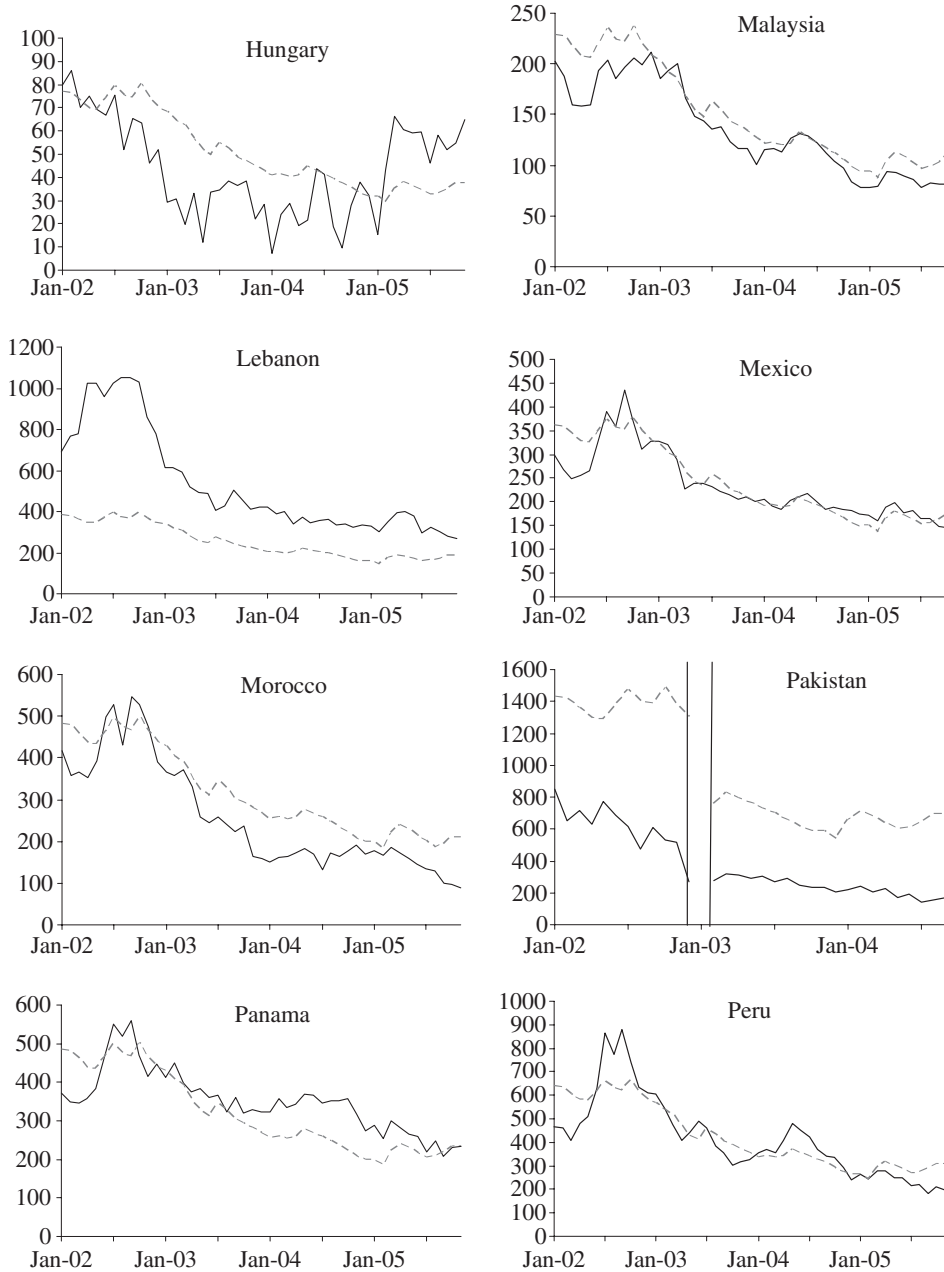

Fig. 3 *Continued*
Note. The solid line indicates the EMBI data; the dotted line indicates predictions using HY and 10YT.

Latin American countries that underwent severe episodes of capital account reversals in the period considered. Second, the predictions highlight the explanatory power of the two exogenous variables: despite short-term, transitory swings due to country-specific episodes, spread movements closely reflect these three variables and eventually

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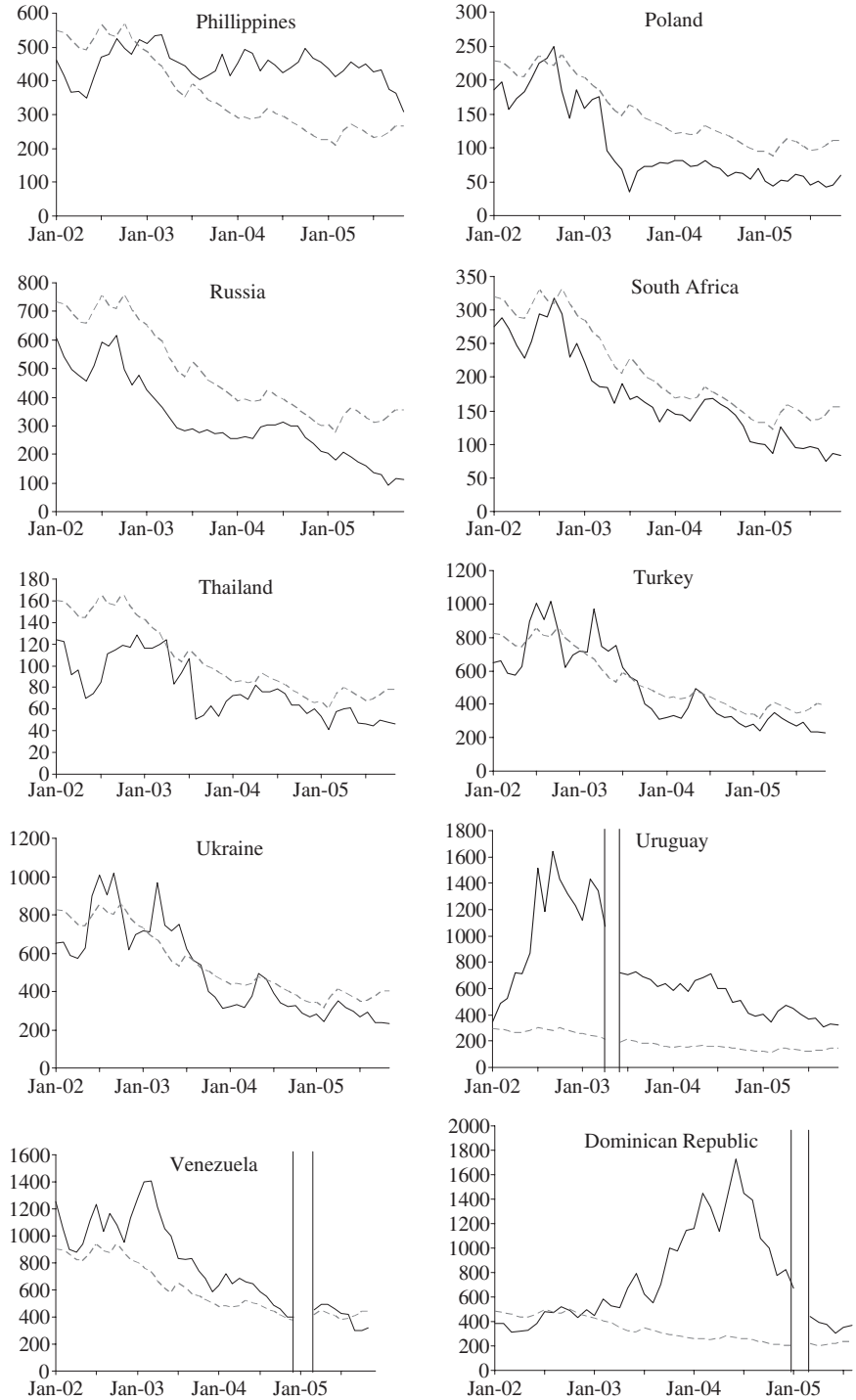

Fig. 3 *Continued*

Note. The solid line indicates the EMBI data; the dotted line indicates predictions using HY and 10YT.

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converge to levels that are largely explained by them. Third, although many of these episodes may have coincided with (and possibly been facilitated by) a deterioration in those global factors, they far exceed the variability of the latter, suggesting that some country-specific ingredient was crucially in place at the time. Brazil is a case in point: the financial turmoil of late 2002 associated with the uncertainty surrounding the election and the transition to a new government, which was clearly independent of the evolution of global factors; once over, spreads rapidly converged to their long-run levels. An even more stark example is provided by Argentina, where the model predicts a spread of about 500 right after the exchange was concluded – very close to the one actually realised (González-Rozada and Levy Yeyati, 2006).

3. Final Remarks

Our tests attempted to estimate the variability explained by global factors but adopted an agnostic approach to country-specific factors, which pose non-trivial empirical problems. Most available country-specific controls are sampled at too low frequency and, even when they are available on a daily basis like credit ratings, are likely to be endogenous. Therefore, an accurate decomposition of emerging market spreads volatility into their systemic and idiosyncratic determinants is still a pending task.

This does not detract, however, from the validity of our main result: global factors explain a large part of the (substantial) volatility of emerging market spreads. Besides its immediate implications for the forecast of the borrowing cost (and, as a result, the fiscal sustainability) of emerging economies, this finding contributes to the debate about the nature of emerging market stability, specifically on the degree of exogeneity in the determination of the highly volatile borrowing costs faced by emerging economies – a major source of financial distress in the recent past.

By documenting the relevance of external factors, the article suggests that the policy focus should be placed on the aspects that make a country more or less resilient to sudden changes in the external context, thereby emphasising debt vulnerability over debt sustainability. Specifically, it highlights the role of country-specific fundamentals as determinants of the *exposure* to external shocks that ultimately drive the evolution of spreads – rather than as determinants of spreads themselves, as proposed by the standard view of debt sustainability. Moreover, to the extent that credit ratings represent a summary measure of the relevant country-specific factors, the finding that they respond to spreads more than they influence them is again in line with the view that the time variability of spreads is largely associated with external factors.

While this does not deny the influence of fundamentals on spreads (particularly on their long-run levels, on which the article is silent), it suggests that, in addition to improving macro fundamentals, emerging economies need to devise mechanisms to reduce their exposure to global factors, as financial integration brings contagion from other emerging economies and, most notably, from the rest of the developed world. Ultimately, in the absence of a concerted effort – both by emerging and developed countries – to reduce these undesired effect of integration, newcomers, like infants, had better take their shot well in advance.

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Appendix

Table A1

Variable Definitions and Sources

Name	Description	Source
Embi	JP Morgan EMBI global index blended spread, in bps	Datastream
HY	CSFB high yield global index, USD, long term debt, in bps	Bloomberg
10YT	US Treasury notes, 10 year constant maturity yield, bps	U.S. Treasury
Rating	S&P rating, long term debt, end of period, foreign currency	S&P
VIX	CBOE Volatility Index	CBOE
Rating + outlook	S&P rating augmented using the S&P outlook. Long term debt, end of period, foreign currency	S&P
Rating (HY)	Weighted average rating of issues included in JP Morgan high yield global index	JP Morgan

Table A2

Countries and Periods Covered

	Monthly			Weekly		
Country	Obs	Begins	Ends	Obs	Begins	Ends
Argentina	101	12/31/93	11/30/05	307	7/4/96	11/24/05
Brazil	132	12/31/94	11/30/05	493	7/4/96	11/24/05
Bulgaria	85	11/30/98	11/30/05	368	11/26/98	11/24/05
China	141	3/31/94	11/30/05	493	7/4/96	11/24/05
Chile	79	5/31/99	11/30/05	341	6/3/99	11/24/05
Colombia	106	2/28/97	11/30/05	458	3/6/97	11/24/05
Croatia	89	1/31/97	5/31/04	388	1/23/97	6/24/04
Dominican Republic	45	11/30/01	11/30/05	189	12/6/01	11/24/05
Ecuador	64	8/31/00	11/30/05	276	8/31/00	11/24/05
Egypt	53	7/31/01	11/30/05	228	8/2/01	11/24/05
El Salvador	44	4/30/02	11/30/05	189	5/2/02	11/24/05
Hungary	83	1/31/99	11/30/05	358	2/4/99	11/24/05
Indonesia	19	5/31/04	11/30/05	80	6/3/04	11/24/05
Korea	124	12/31/93	3/31/04	409	7/4/96	4/29/04
Lebanon	92	4/30/98	11/30/05	398	4/30/98	11/24/05
Malaysia	110	10/31/96	11/30/05	476	10/31/96	11/24/05
Mexico	144	12/31/93	11/30/05	493	7/4/96	11/24/05
Morocco	93	3/31/98	11/30/05	406	3/5/98	11/24/05
Pakistan	39	6/30/01	11/30/05	168	7/5/01	11/24/05
Panama	107	1/31/97	11/30/05	464	1/23/97	11/24/05
Peru	96	12/31/97	11/30/05	417	12/18/97	11/24/05
Philippines	96	12/31/97	11/30/05	415	1/1/98	11/24/05
Poland	126	6/30/95	11/30/05	493	7/4/96	11/24/05
Russia	73	12/31/97	11/30/05	317	1/1/98	11/24/05
South Africa	132	12/31/94	11/30/05	493	7/4/96	11/24/05
Thailand	103	5/31/97	11/30/05	445	6/5/97	11/24/05
Tunisia	43	5/31/02	11/30/05	184	6/6/02	11/24/05
Turkey	114	6/30/96	11/30/05	493	7/4/96	11/24/05
Ukraine	48	12/31/01	11/30/05	207	12/27/01	11/24/05
Uruguay	54	5/31/01	11/30/05	235	5/31/01	11/24/05
Venezuela	142	12/31/93	11/30/05	487	7/4/96	11/24/05

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